Half-life, 4, 86
of actinide elements, 439t
of alpha decay, 155, 157, 158f
in body, 479
decay energy and, 119
metastable and, 32–33
notation for, 6–7
of promethium, 79
for spontaneous fission, 315, 334f
Halide volatility, 709
Hassium (Ha), 440t, 442
Head end plant, 710–713, 711f
Health chemists, 489
Health physicists, 446, 489
Heat conductivity, of fuel, 692, 693f
Heat of formation (ΔH), 70–71
molecular stability and, 72
Heavy water
neutron moderation with, 606t, 607
production of
with chemical exchange, 55, 56f
cryogenic distillation, 60
with electrolysis, 56
Heisenberg uncertainty principle, 120–121
Helium (He)
binding energy for, 72
burning to iron, 358–361, 359f, 360f
cloud chamber pictures of, 15, 16f
hydrogen fusion into, 347, 354–358, 356f, 356t, 357t
isotopic data for, 34t
mass defect of, 72
Helium cooled and graphite-moderated reactor (HTTR), 663–664
Helium flash, 345f, 358
Hertzsprung–Russell (HR) diagrams, 344–345, 345f
Heterogeneous reactors, 600
HF, *see* Hindrance factor
hfs, *see* Hyperfine spectra
High-level liquid wastes (HLLWs)
solidification of, 737f, 739–741, 739f, 741t
tank storage of, 729f, 730–731
High-level wastes (HLWs)
beneficial utilization of, 748–749
deposition in geologic formations, 741–747
repository projects, 740f, 743f, 747
waste conditioning before, 740f, 743f, 745–746
final treatment of, 731–738
dispersion into sea and air, 730f, 731–734, 733f
disposal into space, 737–738
partitioning of, 733f, 734f, 735–736, 736f
remote deposition, 738
transmutation, 738
High-purity germanium (HPGe) detectors, 268, 268f
efficiency curves for, 281, 282f
typical peak from, 282–283, 282f
Hindrance factor (HF), 157, 159–161
HLLWs, *see* High-level liquid wastes
HLWs, *see* High-level wastes
Hold-back carrier, 550, 570
Hole theory, 23
Homogeneous reactors, 600
Hot atom chemistry, 531
Hot atom reactions, 531–532
Hot fusion, 429
HPGe detectors, *see* High-purity germanium detectors
HR diagrams, *see* Hertzsprung-Russell diagrams
HTTR, *see* Helium cooled and graphite-moderated reactor
Hubble constant, 345–346
Hubble law, 345–346
Hydrogen (H)
fusion to helium, 347, 354–358, 356f, 356t, 357t
isotope effect in, 41–42
isotopic data for, 34t
isotopic line shifts for, 42–43
in nuclear reactor water, 222
optical emission spectra for, 42–43
in water cooled reactors, 668
Hydrogen bombs, 648–649, 650f
tritium from, 379
Hydrolysis, of actinide elements, 766f, 767–769
Hydroxyl radical, 223
Hyperfine spectra (hfs), 148–149, 149f

I

Immunoassay, radioactive tracers in, 575
INAA, *see* Instrumental NAA
Industrial radiation processing, 234–235
Inelastic scattering, 297, 306–307

Inertial confinement, 640–641
Injector, 495
Inner van Allen belt, 377
Inorganic nonmetallic compounds
  radiation effects on, 217–219, 218f, 219f
  scintillation detectors for, 270
Instrumental NAA (INAA), 558–561, 559t, 560f, 562t
Internal conversion, 86–87, 99–102, 101f
  Auger electrons compared with, 106
Internal photoelectric process, 106
Internal quantum number, 148
Internal radiation, 480t
Internal radionuclides, radiation therapy with, 586
International Nuclear Event Scale, 628t, 638
International Thermonuclear Reactor Experiment (ITER), 641–642, 642f, 643t
Intrinsic conductivity, 262
Intrinsic detectors, 268, 268f
Intrinsic rotation, 130–131
Ion chambers, 255–258, 256f
Ion exchange
  of radioactive tracers, 554–555
  radioactive tracers in studies of, 571
  radionuclide separation with, 539–540, 540f
Ion pairs, 164–166
  formation per range of radiation, 173, 174f, 175
  in ionization chamber, 254, 255f
Ion sources, 36–37, 36f
  for particle accelerators, 494–497, 495f
Ion vacancies, 217–218, 218f
Ionic radius, of actinide elements, 430t
Ionization, with beta particles, 175–176, 176f, 177f
Ionization chambers, 88–89, *see also* Parallel plate ionization chamber
Ionizing radiation, 164
  discovery of, 2
  electron production and, 164–166
Ions
  in electric and magnetic fields, 35–36, 35f
  specific charge of, 36
Iron (Fe)
  helium burning to, 358–361, 359f, 360f
  LET and oxidation of, 224, 224f
Irradiation
  fast radiochemical separations after, 534–542
    autobatch procedures, 535–537, 536f, 537f
    ^{11}C-labelled compound production, 535
    mass separator procedures, 541f, 542
    on-line procedures, 537–540, 538f–541f
  radionuclide production with, 514–515
    product specifications, 526–529
    second-order reactions in, 518–523, 519f
    target considerations, 523–526
    yields in, 516–518, 517f
  recoil separations after
    hot atom reactions, 531–532
    Szilard–Chalmers process, 532–534
    target recoil products, 529–531, 530f
Island of stability, 428
Isobar, 32
Isobar parabolas, 77–79, 77f–78f
  for promethium, 79, 80f
  for technetium, 80–82, 81f
ISOLDE, *see* Isotope Separation On Line Development
Isomer shift, 196–198, 196f, 199f
Isomeric transition, 98
  daughter recoil in, 106
Isomers, 32, 98
Isotope charts, 104, 104f
Isotope dilution
  analysis of, 556–558, 557f
  mass spectrometer for, 38–40
Isotope effect, 41–42
  in chemical equilibrium, 43–49
    in biological systems, 49
    isotopic ratio, 47
    kinetic energy and temperature, 44–45, 45f
    for paleotemperatures, 47–49, 48f
    partial partition functions, 45–46
    partition function, 43–44
  in chemical kinetics, 49–52, 50f
Isotope exchange, 43
Isotope Separation On Line Development (ISOLDE), 541f, 542

Isotopes, 32
abundances of, 349
of actinium, 410
of americium, 422
atomic mass and, 33, 35
in atomic weight, 34
of berkelium, 423
of californium, 423
of curium, 422
discovery of, 4
mass spectrometer and, 35–37, 35f–36f
of neptunium, 418–419
neutron-deficient, 514
neutron-rich, 514
notation for, 6
optical emission spectra for, 42–43
physicochemical differences for, 41–43
of plutonium, 419, 707t
production of, general considerations, 514–515
of protactinium, 412
relative abundance of, 35
separation processes for, 52–62
chemical exchange, 55–56, 56f
continuous ion-exchange, 60
cryogenic distillation, 60
electrolysis, 56
electromagnetic, 58
equilibrium processes, 52
gas centrifugation, 59–60, 59f
gaseous diffusion, 57–58
multistage, 52–55, 53f
photoionization and photoexcitation processes, 61–62
rate processes, 52
stationary-walled gas centrifugation, 56f, 60–61
of some elements, 8t–11t
of thorium, 410–412
of uranium, 413
Isotopic exchange, 548
Isotopic ratio, 34
chemical equilibrium and, 47
isotope dilution for, 38–40
in nature, 41
of oxygen, 41, 42f
ITER, *see* International Thermonuclear Reactor Experiment

J

JET machine, *see* Joint European Torus machine
Joint European Torus (JET) machine, 641–642, 642f, 643t

K

K, *see* Kerma
K–Ar method, 393–394, 395f
Kerma (K), 215
Kinetic energy
in bimolecular reaction, 51
of fission fragments, 331–333
of fission products, 331
of ions, 35
of light, 19
most probable, 44–45
rest and relativistic mass in, 92–93, 93f
temperature and, 44–45, 45f
K-meson, properties of, 21t
Knock-on-cascade process, 325–326
Koino matter, 23

L

l, *see* Azimuthal quantum number, *see* Orbital quantum number
Labelled compounds, 527–528
^{11}C-labelled compound production, 535
hot atom reactions in synthesis of, 531–532
Large organ doses, 446
Large radiation dose
epidemiological effects at, 460
mental retardation, 465–466
radiation sickness, 460–461, 461t
radiation therapy, 462–465, 461t
stochastic cancer induction, 462f–463f, 465f, 465
regularities of, 450t, 458f–459f, 459–465
Lattice vacancies, 217–218, 218f
Lawrencium (Lr), 424–427
Lawson limit, 640
Lead (Pb)
bremsstrahlung radiation in, 176–177
dose build-up factors in, 189–190, 190f
found with uranium and thorium, 41
Leptons
classification of, 28t
mass classification of, 20
in nuclear reactions, 27